

Improve One Cause

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Test Edge Cases and Fairness

Without opening the old lesson, explain one key idea from Test Edge Cases and Fairness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Build a Paper Prototype

Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Find a Real Problem

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Check the repair and earlier behaviour for Improving and checking regressions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Improving and checking regressions. Proposed explanation: “One repaired example proves improvement everywhere”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Audit a different proposed repair

Equipment catalogue E lists E1 first with capacity 4 and E2 second with capacity 8. Valid whole group sizes are 1 to 10. Choose the first capacity at least the group size, or no match; invalid input needs clarification. Old U0 uses strictly greater capacity and stops at the first fit. Proposed U1 includes equality but keeps searching and returns the last fit. Independently compare U0, U1 and the requirement for groups 8 and 3. Identify the repair and regression, propose U2 and give one validation check. No booking occurs.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Version A makes four errors. Version B makes one after lighting, labels and rules all change. What can you conclude, and how would you isolate the effect of lighting?

6. Solve the given case

The team changes lighting, labels and rules together; errors fall from four to one.

Can it identify the cause of improvement?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Dim photos fail. You change the lighting and all decision rules together; performance improves. Can you tell which change helped? Design a clearer comparison.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 35 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Create sorter version 2 with one evidence-linked change and a before/after test table.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Version A makes four errors. Version B makes one after lighting, labels and rules all change. What can you conclude, and how would you isolate the effect of lighting?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Write the change record

Use S0's failure on 10 Yes with catalogue B version 1. Record the expected and incorrect outputs, the faulty time comparison, the exact S1 correction and the other conditions that must remain. Explain what old evidence to preserve.

My reasoning and evidence

13. W2 Check the repair and earlier behaviour

Plan and operate the S1 repair and regression set described in the lesson, using the appropriate catalogue version for each case. Record expected and observed results separately with reasons and traces. Summarise full passes over all attempted cases, list unrun cases separately and identify any remaining failures. Do not replace observed mistakes with worked answers.

My reasoning and evidence

14. W3 Reject a repair that breaks quiet requests

T1 accepts equal times but omits the quiet check. Explain why it passes 10 Yes yet fails 9 Yes. Use S0's earlier correct result for 9 Yes to justify the word regression. What should change in T1, and which two cases should be rerun first?

My reasoning and evidence

15. W4 Audit a different proposed repair

Equipment catalogue E lists E1 first with capacity 4 and E2 second with capacity 8. Valid whole group sizes are 1 to 10. Choose the first capacity at least the group size, or no match; invalid input needs clarification. Old U0 uses strictly greater capacity and stops at the first fit. Proposed U1 includes equality but keeps searching and returns the last fit. Independently compare U0, U1 and the requirement for groups 8 and 3. Identify the repair and regression, propose U2 and give one validation check. No booking occurs.

My reasoning and evidence

16. W5 Protect the meaning of final evaluation

After seeing a reserved final case fail, a team edits its candidate, reruns that same case and calls the result “a successful unseen test of the new version”. Explain what is wrong and what should be recorded instead. State what must be saved before an evaluation begins.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Improve One Cause

A fictional quiet-activity chooser has a boundary bug and a proposed repair.

A: quiet Yes, 8 minutes; B: quiet No, 5 minutes. Request quiet required Yes.

U0 accepts only duration < available time. U1 fixes $\leq$ but accidentally removes the quiet requirement.

Improve One Cause

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Boundary failure

At 8 minutes, U0 rejects A and rejects noisy B: none.



2. Expected repair

Correct logic must allow A when time equals 8.



3. Regression case

At 6 quiet-required minutes, correct output is none.



4. Bad revision

U1 selects noisy B at 6; fixing equality introduced a new failure.

CHANGE ONE THING

Use U2, which changes $<$ to $\leq$ and keeps the quiet check.

Explain the changed result using the matching lesson.

Explain why “our repaired version now picks something” is not enough to show success at 6.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Maintain sorter regression tests and compare complete version traces, retaining new failures rather than hiding them.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Maintain sorter regression tests and compare complete version traces, retaining new failures rather than hiding them.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional quiet-activity chooser has a boundary bug and a proposed repair.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? U1 selects noisy B at 6; fixing equality introduced a new failure.

1. Bad revision
2. Regression case
3. Expected repair
4. Boundary failure

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Bad revision U1 selects noisy B at 6; fixing equality introduced a new failure. A repair must preserve the full contract, not just make one failing example pass. The quiet condition and duration comparison are separate requirements. Changing only the inequality targets the recorded cause; removing the quiet check changes another behavior. Save previously passing cases to reveal such regressions.

Practice 2: explain the deciding evidence

Changed-case practice: Use U2, which changes $<$ to $\leq$ and keeps the quiet check. Which conclusion is supported by the supplied facts?

1. At 8 return A; at 6 return none.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: At 8 return A; at 6 return none. Equality is repaired while B remains ineligible because it is noisy.

Repair a missing building block

Review Test Edge Cases and Fairness (ai-1/34). Goal: Choose tests just below, at and above a decision boundary. Short practice: Which single boundary run most directly detects the strict-comparison bug, and why?. Return to this week and explain what you changed.

Review Build a Paper Prototype (ai-1/32). Goal: Build a runnable paper prototype with explicit instructions and visible state. Short practice: What should appear in the test log before and after the repair?. Return to this week and explain what you changed.

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied evidence without inventing its cause. Short practice: Write a one-sentence project goal and say what evidence would count as a failure.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
